MT 3 Review

- Tuesday (11/10)
- 8-9 PM ← Show up at 7:45 - 7:50
- Lectures 19-28 [See Below]
- WTHR 200 ← Main Floor NOT Balcony

Additional Office Hours this week

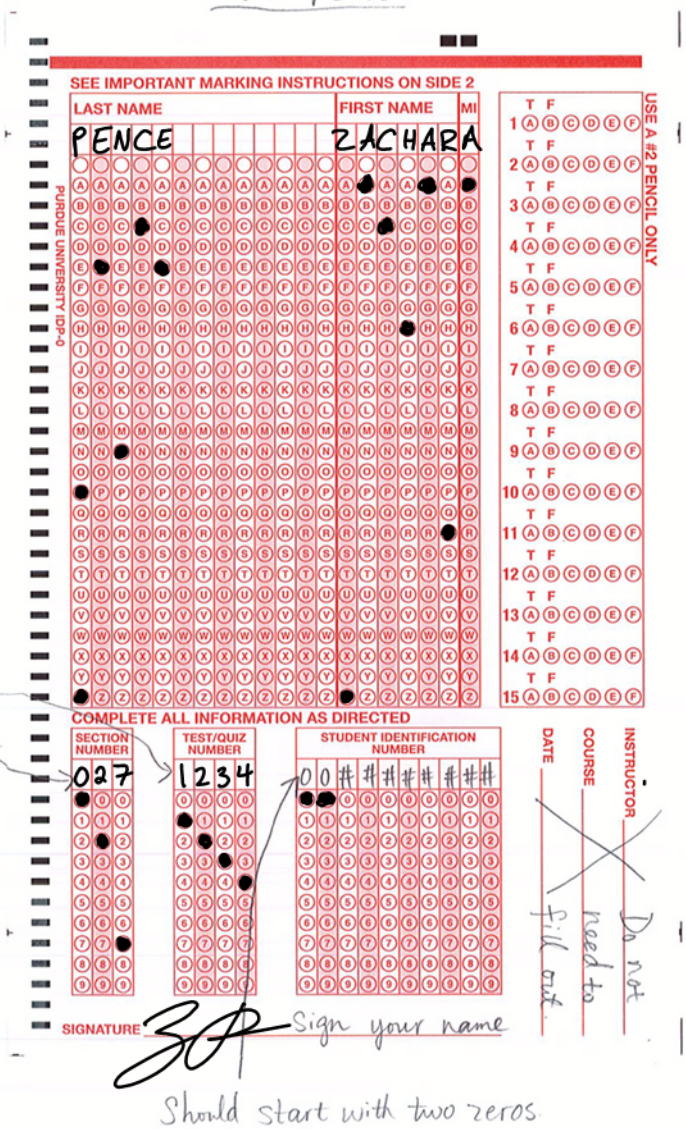
- Tuesday, 3:30 - 4:30 in my office (MATH 615)
- Usual office hours (Tues 2:30 - 3:30, MATH G175)

- No class Wednesday



Make Sure to Bring:

- Pencil / Erasers
- PUID
- Calculator TI 30XA Single Line Only



Midterm 3 Review

NOTE: These are just review problems of Lectures 19-28; these are not necessarily representative of the problems of the exam. The exam can (and most likely will) have different problems.

Topics of Exam 3:

- **Lecture 19:** Absolute Extrema on an Interval
- **Lecture 20:** Graphical Interpretation of Derivatives
- **Lecture 21:** Limits at Infinity
- **Lecture 22:** A Summary of Curve Sketching
- **Lecture 23-25:** Optimization
- **Lecture 26-27:** Antiderivatives and Indefinite Integration
- **Lecture 28:** Area and Riemann Sums

Problem 1. *Let M and m be the maximum and minimum of $f(x) = x^3 - 3x^2 + 3$ on the interval $[1, 4]$, respectively. What is the value of $M + m$?*

$f(x) = x^3 - 3x^2 + 3$

$f'(x) = 3x^2 - 6x$ set 0

$3x(x-2) = 0$

$x = 0, 2$

$x \in [1,4]$ $\Longrightarrow$ $x = 2$

x	1	2	4
$f(x)$	1	-1	19
		min	max

$x^3 - 3x^2 + 3$

$M = 19$

$m = -1$

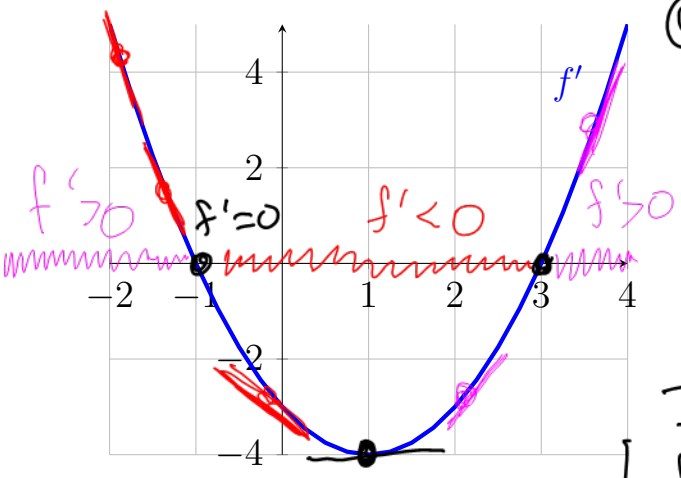
$M + m = 19 - 1 = 18$

Problem 2. *A graph of f' is given below.*

(a) Determine when f is increasing and when it is decreasing.

(b) Determine when f is concave up and when it is concave down.

(c) Locate the positions (x-coordinates) of any relative extrema and inflection points.



@ Conduct 1D Test

Crit Nums: $-1, 3$



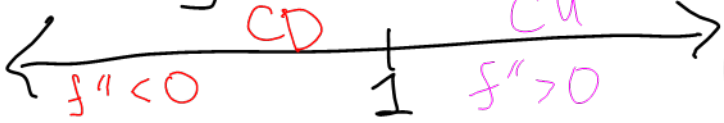
Inc: $(-\infty, -1) \cup (3, \infty)$

Dec: $(-1, 3)$

Rel max: at $x = -1$

Rel min: at $x = 3$

Concavity and IPs



f has an IP at $x = 1$

$\frac{x^2+1}{2-x}$

Problem 3. *Compute* $\lim_{x\to\infty} \frac{x^2+1}{2-x}$ *and* $\lim_{x\to-\infty} \frac{x^2+1}{2-x}$.

$$\lim_{x\to\infty} \frac{x^2+1}{2-x} = \lim_{x\to\infty} \frac{x^2}{-x} = \lim_{x\to\infty} -x = -\infty$$

$$\lim_{x\to-\infty} \frac{x^2+1}{2-x} = \lim_{x\to-\infty} -x = \infty$$

Problem 4. *Use the techniques learned in Lecture 22 to sketch a graph of the function* $y = \frac{x^2}{x+8}$.

$$y = \frac{x^2}{x+8}$$

$\deg(x^2) = 2$ ←

$\deg(x+8) = 1$ ←

Determine slant asy:

$$x+8 \overline{)\, x^2}$$

$x - 8$ ← the quotient is the slant asy.

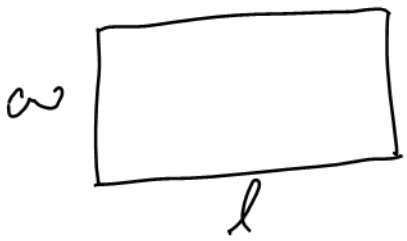
$$-(x^2 + 8x)$$

$$-8x$$

$$-(-8x - 64)$$

$$64$$

Problem 5. *If a rectangle has a fixed perimeter of 40, what is its maximum area?*



Objective: Maximize area

$\max A(l,w) = lw$

Given: $2w + 2l = 40$

$2l = 40 - 2w$

$\boxed{l = 20 - w}$

$A(w) = (20-w)w$

$A(w) = 20w - w^2$

$A'(w) = 20 - 2w \overset{\text{set}}{=} 0$

$20 = 2w$

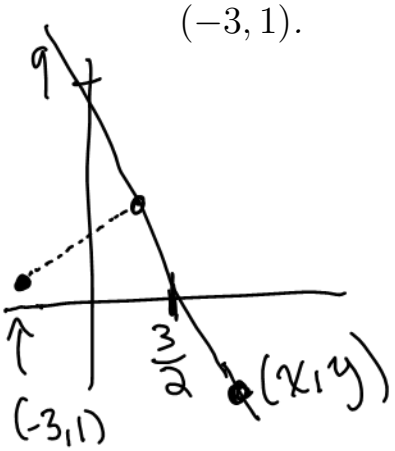
$w = 10$

Need to compute $A(10)$:

$A(w) = (20-w)w$

$A(10) = (20-10)10 = 100 \text{ un}^2$

Problem 6. *Find the point on the line* $6x + y = 9$ *that is closest to the point* $(-3, 1)$.



Obj: Minimize distance from $(-3,1)$

$$d(x,y) = \sqrt{(x+3)^2 + (y-1)^2}$$

It suffices to minimize the fcn

$$S(x,y) = [d(x,y)]^2 = (x+3)^2 + (y-1)^2$$

objective

Given: $6x + y = 9 \Rightarrow y = -6x + 9$

$$S(x) = (x+3)^2 + (-6x+9-1)^2$$

$$S(x) = (x+3)^2 + (-6x+8)^2$$

$$S(x) = x^2 + 6x + 9 + 36x^2 - 96x + 64$$

$$S(x) = 37x^2 - 90x + 73$$

$$S'(x) = 74x - 90 \overset{\text{set}}{=} 0$$

$$x = \frac{90}{74} = \frac{45}{37}$$

We need to find the y coord:

$$y = -6x + 9$$

$$y = -6\left(\frac{45}{37}\right) + 9 = \frac{63}{37}$$

Therefore, the point $\left(\frac{45}{37}, \frac{63}{37}\right)$ is closest to $(-3,1)$

Problem 7. *A particle is moving in a straight line; its acceleration function is given by* $a(t) = \cos t + \sin t$ *for* $t \geq 0$. *Find the position of the particle at time* t *when the particle has an initial position of 0 and an initial velocity of 5.*

$s''(t) \rightarrow a(t) = \cos t + \sin t$

$s'(0) \rightarrow v(0) = 5$

$s(0) = 0$

$a(t) = \cos t + \sin t$

$v(t) = \sin t + (-\cos t) + C$

$v(0) = 5$

$5 \overset{\text{should}}{=} v(0) = 0 - 1 + C$

$5 = -1 + C$

$C = 6$

$v(t) = \sin t - \cos t + 6$

$s(t) = -\cos t - \sin t + 6t + D$

$s(0) = 0$

$0 \overset{\text{should}}{=} s(0) = -1 - 0 + 0 + D$

$D = 1$

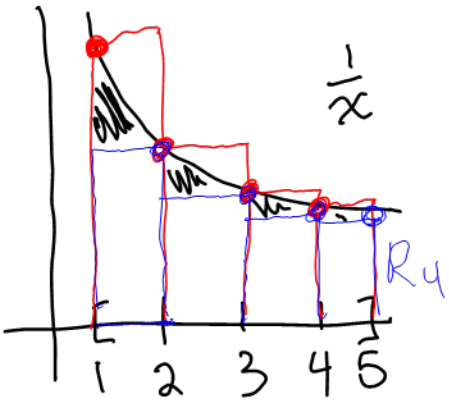
$s(t) = -\cos t - \sin t + 6t + 1$

Problem 8. *We estimate the area underneath the graph of* $f(x) = \frac{1}{x}$ *on the interval* $[1, 5]$.

(a) *Compute the left Riemann sum with* 4 *rectangles. Is this an overestimate or an underestimate? Explain why (The exact area is* $\ln 5$*, but you don't need to know that to answer the question).*

(b) *Repeat (a) using the right Riemann sum.*

(c) *Set up (but do not compute) the left Riemann sum using* N *rectangles (where* N *is a positive integer).*



(a) Compute L_4

Width: $\Delta x = \frac{b-a}{N} = \frac{5-1}{4} = 1$

(b) $L_N = \sum_{i=0}^{N-1} f(a + i\Delta x)\Delta x$

$$L_4 = \sum_{i=0}^{3} f(1 + i(1))(1) = \sum_{i=0}^{3} \frac{1}{1+i}$$

$$= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{25}{12} \leftarrow \text{Over estimate}$$

(b) Compute R_4

$$R_N = \sum_{i=1}^{4} \frac{1}{1+i} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{77}{60}$$

↑ Under estimate

(c) Set up L_N and R_N

$$\Delta X = \frac{b-a}{N} = \frac{5-1}{N} = \frac{4}{N}$$

$$L_N = \sum_{i=0}^{N-1} f(a + i\Delta X)\Delta X = \sum_{i=0}^{N-1} \frac{1}{1 + i\left(\frac{4}{N}\right)} \left[\frac{4}{N}\right]$$

$$R_N = \sum_{i=1}^{N} \frac{1}{1 + i\left(\frac{4}{N}\right)} \left[\frac{4}{N}\right]$$